

torch.autograd.function.FunctionCtx.ma

<https://pytorch.org/docs/stable/generated/torch.autograd.function.FunctionCtx>

Description:

Mark given tensors as modified in an in-place operation. This should be called at most once, in either the `setup_context()` or `forward()` methods, and all arguments should be inputs. Every tensor that's been modified in-place in a call to `forward()` should be given to this function, to ensure correctness of our checks. It doesn't matter whether the function is called before or after modification.

Function Signature

```
FunctionCtx.mark_dirty(*args)[source]#
```

Examples::

Code Examples

```
>>> class Inplace(Function):
>>>     @staticmethod
>>>     def forward(ctx, x):
>>>         x_numpy = x.numpy() # x_numpy shares storage with x
>>>         x_numpy += 1
>>>         ctx.mark_dirty(x)
>>>         return x
>>>
>>>     @staticmethod
>>>     @once_differentiable
>>>     def backward(ctx, grad_output):
>>>         return grad_output
>>>
>>> a = torch.tensor(1., requires_grad=True,
>>> dtype=torch.double).clone()
>>> b = a * a
>>> Inplace.apply(a) # This would lead to wrong gradients!
>>>                 # but the engine would not know unless we
>>> mark_dirty
>>> b.backward() # RuntimeError: one of the variables needed for
>>>              gradient
>>>              # computation has been modified by an inplace
>>> operation
```

Detailed Documentation

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